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Original Research Article

Estimating the serum folate concentration that corresponds to the red blood cell folate concentration threshold associated with optimal neural tube defects prevention: A population-based biomarker survey in Southern India

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A B S T R A C T

Background: RBC folate concentrations are monitored at the population level, with a recommended threshold for optimal neural tube defect (NTD) prevention. A corresponding threshold for serum folate has not been established.

Objectives: This study aimed to estimate the serum folate insufficiency threshold corresponding to the RBC folate threshold for NTD prevention and examine how this threshold is modified by vitamin B_{12} status.

Methods: Participants were women (15–40 y; not pregnant or lactating; $n = 977$) from a population-based biomarker survey in Southern India. RBC folate and serum folate were measured via microbiologic assay. RBC folate deficiency (<305 nmol/L) and insufficiency (<748 nmol/L), serum vitamin B_{12} deficiency (<148 pmol/L) and vitamin B_{12} insufficiency (<221 pmol/L), elevated plasma MMA (>0.26 μmol/L), elevated plasma homocysteine (>10.0 μmol/L), and elevated HbA1c (≥6.5%) were evaluated. Bayesian linear models were used to estimate unadjusted and adjusted thresholds.

Results: Compared with adequate vitamin B_{12} status, the estimated serum folate threshold was higher in participants with serum vitamin B_{12} deficiency (72.5 vs. 28.1 nmol/L) or vitamin B_{12} insufficiency (48.7 vs. 24.3 nmol/L) and elevated MMA (55.6 vs. 25.9 nmol/L). The threshold was lower in participants with elevated HbA1c (HbA1c ≥6.5% vs. <6.5%; 21.0 vs. 40.5 nmol/L).

Conclusions: The estimated serum folate threshold for optimal NTD prevention was similar to previous reports (24.3 vs. 25.6 nmol/L) among participants with sufficient vitamin B_{12} status. However, this threshold was more than 2-fold higher in participants with vitamin B_{12} deficiency and substantially higher across all indicators of insufficient vitamin B_{12} status (<221 pmol/L, elevated MMA, combined B_{12}, impaired vitamin B_{12} status), and lower in participants with elevated HbA1c. Findings suggest a serum folate threshold for NTD prevention may be possible in some settings; however, it may not be appropriate in populations with high prevalence of vitamin B_{12} insufficiency. *Am J Clin Nutr* 2023;xx:xx–xx.

This trial was registered at https://clinicaltrials.gov as NCT04048330.

Keywords: periconceptional, vitamin B_{12}, folate, nutrition, India

Abbreviations: AGP, α-1 acid glycoprotein; BRINDA, Biomarkers Reflecting Inflammation and Nutritional Determinants of Anemia; cB_{12}, combined B_{12}; Hb, hemoglobin; Hcy, homocysteine; LOD, limit of detection; 5-MTHR, 5-methyltetrahydrofolate; NTD, neural tube defect; TF, trunk fat; THF, tetrahydofolate; WBF, whole body fat; WC, waist circumference.

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Introduction

Neural tube defects (NTDs) are structural birth defects that result from failure of neural tube closure during embryonic development [1]. Over 260,000 NTDs occur globally each year (1–80 NTDs per 10,000 births), with the highest burdens in low and middle-income countries [2,3]. India is estimated to have among the highest burdens of NTDs [2,4,5].

Folate is required for DNA synthesis and cell division and is critical during periods of rapid growth and development [6]. The association of low maternal periconceptional folate status with NTD risk has been established [7,8]. Evidence from randomized trials [9,10] and a large community-based study [11] demonstrated periconceptional folic acid supplementation reduced NTD occurrence. These findings informed dietary recommendations for women of reproductive age in the United States [12] and folic acid fortification in over 70 countries [13]. Higher red blood cell (RBC) folate concentrations have been associated with lower risk of NTDs in observational studies [8,14] and Bayesian modeling analyses [15], and fortification of flour with folic acid has been consistently associated with reductions in NTDs at the population level [13,16–21].

The World Health Organization (WHO) recommends monitoring RBC folate concentrations in women of reproductive age at the population level. A threshold of RBC folate ≥906 nmol/L, identified in an observational study and replicated through a Bayesian analysis, is recommended for optimal NTD prevention [15,22]. However, population-level RBC folate data is limited. Most national surveys evaluate serum or plasma folate, due in part to the reduced cost and complexity [6]. Of the 38 national surveys to date reporting folate status, 18 included RBC folate, of which 10 used the recommended microbiologic assay [23]. RBC folate reflects average folate intake during the past ~90 to 120 d and has been consistently associated with NTD risk whereas serum/plasma folate reflects recent folate intake [6]. To our knowledge, no studies have evaluated associations between periconceptional serum/plasma folate and NTD risk.

A Bayesian analysis estimated a plasma folate threshold corresponding to the RBC folate threshold for optimal NTD prevention [24]. The findings identified a threshold of 27.2 nmol/L for optimal NTD prevention. This threshold was modified by vitamin B_{12} and overweight status; the authors concluded it would not be applicable in populations with a high burden of vitamin B_{12} deficiency [24]. Vitamin B_{12} is required for the conversion of 5-methyltetrahydrofolate (5-MTHF) to tetrahydrofolate (THF) and retention of folates in proliferating cells such as the developing RBC during erythropoiesis [6]. To our knowledge, no studies have estimated a serum or plasma folate threshold for NTD prevention in a population with high prevalence of vitamin B_{12} deficiency.

The objectives of this analysis were to estimate the serum folate threshold corresponding to the RBC folate threshold for optimal NTD prevention using data from a population-based biomarker survey in Southern India [25,26] and to evaluate how this threshold was modified by vitamin B_{12} status (e.g., vitamin B_{12} <148 pmol/L, <221 pmol/L), as well as other sociodemographic (e.g., age), nutritional (e.g., anemia, iron deficiency), and metabolic (e.g., BMI, HbA1c) characteristics.

Methods

Population

This analysis was conducted using data from a population-based biomarker survey in Southern India (NCT04048330). Participants were women 15 to 40 y, who were not pregnant or lactating. The design and findings from this study have been previously described [25,26]. Briefly, a census was conducted for all households in a 50 km^2 catchment area; all rural households (n = 3124) and a random sample of urban households (n = 1000) were selected for screening and recruitment.

Ethics

The study protocol was reviewed and approved by the Institutional Review Board at Cornell University in the United States and the Institutional Ethics Committees at Arogyavaram Medical Centre and St. John's Research Institute in India [25,26]. This study received clearance from the Indian Council of Medical Research Health Ministry Screening Committee and was registered at clincialtrials.gov (NCT04048330). The protocol for the biomarker survey was reviewed in accordance with the US Centers for Disease Control and Prevention human research protection procedures and was determined to be a nonresearch, routine surveillance activity. Written informed consent (≥18 y) or assent (15 to <18 y) was obtained from all participants before the start of data collection.

Data collection

Data was collected at Arogyavaram Medical Centre on electronic tablets, using standardized procedures [25,27]. Data collection included sociodemographic, dietary, anthropometric, and clinical information, and biological specimens (i.e., blood, saliva, urine). Anthropometric measurements included weight (via Seca 874 DR Doctor's scale to the nearest 0.01 kg), height (via Shorr HeightLite stadiometer to the nearest 0.1 cm), and hip and waist circumference (WC) (via tape measure to the nearest 0.1 cm). BIA (BC-148 MA; Tanita Corporation; 8 electrode) was conducted among adults (≥18 y).

Laboratory analyses

Venous blood samples were collected by a trained phlebotomist, processed, and stored using standardized protocols [25]. Whole-blood samples were analyzed for hemoglobin (Hb) via automated Coulter counter (Coulter HMX) and HbA1c via nephelometry (Mispa i2 Specific Protein Analyzer, Agappe Diagnostics). Plasma, serum, and RBCs were processed and stored at -80ºC or less until batch analysis at the end of data collection. RBC folate and serum folate concentrations were measured using the WHO-recommended microbiologic assay [28]. Serum total vitamin B_{12} and serum ferritin concentrations were assessed via chemiluminescence on the Roche E411 analyzer (Roche Diagnostics). Soluble transferrin recepter (sTfR), serum C-reactive protein (CRP), and α-1 acid glycoprotein (AGP) concentrations were analyzed via immunoturbidimetry on the Roche COBAS Integra 400 plus analyzer (Roche Diagnostics).

Plasma methylmalonic acid (MMA) and plasma homocysteine (Hcy) were assessed by gas chromatography-mass spectrometry in a methylchloroformate derivatization (GC-MS, 5975, Agilent Technologies). The concentration ranges and mean coefficients of variation for quality controls measured in each assay were: serum folate (low: 5.6–11.3 nmol/L, 7.43%; high: 20.6–41.9 nmol/L, 7.12%), RBC folate (low: 127–300 nmol/L, 8.8%; high: 322–595 nmol/L, 7.5%), vitamin B_{12} (low: 296–445 pg/mL, 7.5%; medium: 322–595 pg/mL, 6.1%; high: 607–834 pg/mL, 4.5%), serum ferritin (low: 5.3–88.0 ng/mL, 2.4%; medium: 128–196 ng/mL, 3.7%; high: 276–424 ng/mL, 1.5%), CRP (low: 8.5–10.4 mg/L, 2.8%; high: 26.7–32.6 mg/L, 2.6%), sTfR (low: 2.0–2.5 mg/L, 3.5%; high: 6.6–8.1 mg/L, 3.1%), AGP (low: 0.4–0.6 g/L, 2.9%; high: 0.7–1.0 g/L, 2.2%), MMA (in-house pooled

plasma: 0.4–0.5 μmol/L, 3.7%), Hcy (in-house pooled plasma: 6.8–8.1 μmol/L, 2.8%).

Definitions of variables

Folate deficiency was defined as RBC folate <305 nmol/L (i.e., risk of macrocytic anemia), and folate insufficiency was defined as RBC folate <748 nmol/L, the recommended calibrator-adjusted equivalent of the RBC folate threshold for optimal NTD prevention [22]. Anemia was defined as Hb <12.0 g/dL, and severe anemia was defined as Hb <8.0 g/dL [29]. Vitamin B_{12} deficiency and vitamin B_{12} insufficiency were defined as vitamin B_{12} <148 and <221 pmol/L, respectively [30]. Elevated MMA was defined as MMA >0.26 and >0.37 μmol/L [30, 31]. Impaired vitamin B_{12} status was defined as total vitamin B_{12} <148 pmol/L and MMA >0.26 μmol/L. A composite indicator of vitamin B_{12} status (i.e., combined B_{12}; cB_{12}) using 3 biomarkers (i.e., vitamin B_{12}, MMA, Hcy) was calculated and categorized (i.e., ≥1.5, -0.5 to <1.5, -1.5 to <-0.5, -2.5 to <-1.5, <-2.5) using methods developed by Fedosov et al [32]. Elevated Hcy was defined as >15.0 and >10.0 μmol/L [6]. Iron deficiency was defined as serum ferritin <15.0 μg/L; iron insufficiency was defined as serum ferritin <20.0 and <25.0 μg/L [33,34]. Body iron index was estimated using an equation proposed by Cook after converting the Roche sTfR data from this study to data equivalent to the original Flowers enzyme-linked immunosorbent assay used in the development of the body iron model (i.e., via the equation Flowers sTfR = 1.5 × Roche sTfR + 0.35 mg/L) [35]. Elevated inflammatory biomarkers were defined as CRP >5.0 mg/L or AGP >1.0 g/L [33]; additional definitions of CRP >3.0 mg/L and >1.0 mg/L [36] were also considered.

BMI was calculated as weight (kg) divided by height (m) squared and was categorized among adults (≥18 y) using WHO criteria (i.e., BMI <18.5, 18.5 to <25.0, 25.0 to <30.0, and ≥30.0 kg/m^2) [37]. Additional anthropometric measurements (e.g., elevated WC, elevated WHR) were categorized among adults (≥18 y) using WHO criteria [38], and elevated whole body and trunk fat percent (%) were defined based on criteria suggested by the American Association of Clinical Endocrinology [39]. HbA1c levels were categorized as <5.7%, ≥5.7% to <6.5%, and ≥6.5%, using American Diabetes Association criteria [40] and also defined as <5.7% and ≥5.7%.

The primary outcomes for the population-based biomarker survey were Hb concentrations, anemia; RBC folate and serum folate concentrations, RBC folate deficiency and insufficiency; and vitamin B_{12} concentrations, vitamin B_{12} deficiency and insufficiency. The current analysis leverages paired RBC folate and serum folate biomarker data to estimate a population-based serum folate insufficiency threshold corresponding to the RBC folate threshold for optimal NTD prevention.

Statistical analyses

Descriptive analyses.

Continuous biomarker variables were natural logarithmically transformed prior to analyses. Biomarkers outside of the assay limits of detection (LOD) were set to half the LOD (below the lower LOD), or 2 times the LOD (above the upper LOD). Hb was adjusted for self-reported smoking status [29]. Serum ferritin concentrations were adjusted for inflammation using Biomarkers Reflecting Inflammation and Nutritional Determinants of Anemia (BRINDA) methods [41]. Continuous variables were described using geometric means and 95% confidence intervals.

Bayesian linear models.

The Bayesian framework was used to develop population-specific Bayesian linear models, to estimate RBC folate concentrations as a function of serum folate concentrations using procedures developed by Crider et al. [15,24]. Briefly, the joint posterior probability distribution for parameters was calculated using Markov chain Monte Carlo simulations. Parameters were estimated using a posterior sample of 10,000, generated by selecting 500,000 samples (i.e., with a post burn-in of 10, 000 samples), and thinning by retaining every 50th sample. Mildly informative priors of N (mean: 0, standard deviation:10) were used for all parameters. For each model, the 10,000 posterior samples of the estimated serum folate insufficiency threshold corresponding to the RBC folate threshold of <748 nmol/L were summarized by reporting the median (50th percentile) and 95% credible interval (i.e., the plausible range of values within which an unobserved parameter value falls with a 95% probability; 2.5th and 97.5th percentiles). The Bayesian credible interval describes which parameters lie in a given probability range. A 95% credible interval is a range of parameters/values that account for 95% of the posterior distribution.

Stratified analyses.

Sociodemographic (e.g., age, parity), nutritional (e.g., anemia, vitamin B_{12} deficiency), and metabolic (e.g., BMI categories, elevated HbA1c) characteristics were considered in stratified analyses, including variables identified in the previous analysis of the plasma folate insufficiency threshold for NTD prevention [24]. Variables were included in models as interaction terms to facilitate estimation of serum folate threshold by strata of interest, and strata were considered substantially different if the 95% Credible Interval of the difference did not include zero.

Multivariable analyses.

Associations between variables considered for adjustment (e.g., sociodemographic, nutritional, and metabolic characteristics) were examined using unsupervised hierarchical clustering methods. Both individual variables and identified clusters (Supplementary Figures 1–5) were considered for inclusion in adjusted models. Multivariable models from the previous Bayesian analysis [24] were also replicated to examine comparability of findings (Supplementary Table 1). Missing data for covariates (<2%) were assumed to be missing at random and estimated jointly with model parameters of interest. The findings for analyses with and without simulation of missing data were similar; for interpretation purposes, results from full-case models are reported.

Sensitivity analyses.

Sensitivity analyses were also conducted within strata, including vitamin B_{12} status (i.e., vitamin B_{12} <148, ≥148, <221, and ≥221 pmol/L).

Statistical analyses were conducted using SAS 9.4 (SAS Institute, Inc). All models were estimated using SAS MCMC. Analyses were reproduced by the Cornell Results Reproduction, R^2 at the Cornell Institute for Social and Economic Research.

Results

Characteristics of the study population

Characteristics of participants in this study are presented in Table 1, and a flow chart of study participants is presented in Figure 1. Overall, 79.3% of participants had RBC folate <748 nmol/L, 41.5% had anemia

TABLE 1
Characteristics of the study population (n = 977)[1].

Variables	n	Geometric Mean (95% Confidence Interval) or n (%)
Sociodemographic		
Age, y	977	28.8 (28.4, 29.3)
15 to <18		49 (5.0)
18 to <26		228 (23.3)
26 to <36		452 (46.3)
36 to 40		248 (25.4)
Biochemical		
RBC folate[2], nmol/L	977	540.5 (526.2, 555.1)
<305		74 (7.6)
<748		775 (79.3)
Serum folate[2], nmol/L	977	16.5 (16.0, 17.0)
<7.0		34 (3.5)
Hemoglobin[3], g/dL	977	11.9 (11.7, 12.0)
<12.0		405 (41.5)
<8.0		29 (3.0)
Serum ferritin[4], μg/L	975	10.6 (9.9, 11.2)
<15.0		599 (61.4)
<20.0		709 (72.7)
<25.0		790 (81.0)
Body iron index[5], mg/kg	975	-0.7 (-1.0, -0.4)
<0.0		506 (51.9)
Serum CRP[6], mg/L	977	1.2 (1.1,1.4)
≤1.0		425 (43.5)
>1.0 to ≤3.0		261 (26.7)
>3.0 to ≤5.0		123 (12.6)
>5.0		168 (17.2)
Serum α-1 acid glycoprotein[7], g/L	975	0.8 (0.8,0.8)
>1.0		216 (22.2)
Serum vitamin B_{12}[8], pmol/L	977	155.9 (150.1, 161.8)
<148		472 (48.3)
148–221		255 (26.1)
≥221		250 (25.6)
Plasma MMA[8], μmol/L	977	0.3 (0.3, 0.3)
>0.26		592 (60.6)
>0.37		432 (44.2)
Impaired vitamin B_{12} status[8]	977	
Vitamin B_{12} <148 pmol/L and MMA >0.26 μmol/L		370 (37.9)
Plasma homocysteine[9], μmol/L	977	6.6 (6.4, 6.8)
>10.0		157 (16.1)
>15.0		65 (6.7)
cB_{12}[8]	977	-0.35 (-0.41, -0.30)
≥1.5		20 (2.1)
-0.5 to <1.5		517 (52.9)
-1.5 to <-0.5		333 (34.1)
-2.5 to <-1.5		100 (10.2)
<-2.5		7 (0.7)
≥-0.5		537 (55.0)
<-0.5		440 (45.0)
Anthropometric		
BMI, kg/m^2	966	22.6 (22.2, 22.9)
BMI[10,11]	917	
<18.5		176 (19.2)
18.5 to <25.0		440 (48.0)
25.0 to <30.0		213 (23.2)
≥30.0		88 (9.6)
Waist circumference[10,12], cm	966	74.6 (73.9, 75.4)
>88.9	917	123 (13.4)
Waist-hip ratio[10,12]	917	0.8 (0.8, 0.8)
≥0.85		191 (20.8)
Metabolic characteristics		
Whole body fat[10,12], %	919	30.6 (30.0, 31.2)
>35		360 (39.2)
Trunk fat mass[10,12], %	919	27.8 (27.0, 28.7)
>35		332 (36.1)
HbA1c[13], %	977	5.4 (5.4, 5.5)
<5.7		732 (74.9)
≥5.7 to <6.5		196 (20.1)
≥6.5		49 (5.0)

[1] Geometric means and 95% Confidence Intervals were generated using Proc Genmod among all participants with RBC folate (n = 977). BRINDA, Biomarkers Reflecting Inflammation and Nutritional Determinants of Anemia; cB_{12}, combined B_{12}.
[2] Cutoffs for RBC folate deficiency (RBC folate: <305 nmol/L) and RBC folate insufficiency (RBC folate: <748 nmol/L)
[3] Hemoglobin values from complete blood count; adjusted for smoking status [29]
[4] Serum ferritin concentrations were adjusted for inflammation via BRINDA methods [41]; cutoffs for iron deficiency (serum ferritin <15.0 μg/L) and iron insufficiency (serum ferritin <20.0 and <25.0 μg/L) from [33,34]
[5] Values are arithmetic mean (95% CI); body iron index (mg/kg) was estimated using Cook's equation after adjusting serum ferritin for inflammation via BRINDA methods [41] and converting sTfR to Flowers Ramco equivalents (Ramco = 1.5 × Roche + 0.35) [35]
[6] CRP cutoffs for inflammation (>5.0 mg/L) and metabolic risk (>3.0 mg/L and >1.0 mg/L) from [33,36]
[7] α-1 acid glycoprotein cutoff (>1.0 g/L) based on WHO criteria [33]
[8] Cutoffs for vitamin B_{12} deficiency (<148 pmol/L) and vitamin B_{12} insufficiency (<221 pmol/L) from (30); cB12 was calculated using vitamin B12, MMA, and homocysteine, and categorized (i.e., ≥1.5, -0.5 to <1.5, -1.5 to <-0.5, -2.5 to <-1.5, <-2.5) using methods developed by Fedosov et al. [32] Elevated vitamin B12 (≥1.5), Adequate vitamin B12 (-0.5 to <1.5), Decreased vitamin B12 (-1.5 to <-0.5) , Possibly deficient (-2.5 to <-1.5) , Probably deficient (<-2.5) Sufficient vitamin B12 (-0.5 to >1.5), Deficient vitamin B12 (<-0.5).
[9] Homocysteine cutoffs (>15.0 and >10.0 μmol/L) from [6]
[10] Among adult participants (>18 y; n = 928)
[11] BMI categories (BMI <18.5, 18.5 to <25.0, 25.0 to <30.0, and ≥30.0 kg/m^2) as defined by the WHO [37]
[12] Anthropometric cutoffs (waist circumference >88.9 cm and waist-hip ratio ≥0.85) were defined based on WHO criteria [38]; elevated whole body (>35%) and trunk fat (>35%) were defined based on criteria suggested by the American Association of Clinical Endocrinology [39]
[13] HbA1c cutoffs (<5.7%, ≥5.7% to <6.5%, and ≥6.5%) were defined using the American Diabetes Association criteria [40]

(Hb <12.0 g/dL), 48.3% had vitamin B_{12} deficiency (<148 pmol/L), and 37.9% had impaired vitamin B_{12} status (vitamin B_{12} <148 pmol/L and MMA >0.26 μmol/L). A total of 16.1% had Hcy >10.0 μmol/L and 25.1% had HbA1c ≥5.7%. In adults (≥18 y), 19.2% were underweight (BMI <18.5 kg/m^2), 48.0% were normal weight (≥18.5 to <25.0 kg/m^2), 23.2% were overweight (≥25.0 to <30.0 kg/m^2), and 9.6% were obese (≥30.0 kg/m^2) based on WHO criteria (Table 1, Figure 2).

Estimated serum folate insufficiency threshold.

The overall estimated median serum folate insufficiency threshold corresponding to the RBC folate threshold for optimal NTD prevention in this population was 37.8 nmol/L (95% credible interval: 33.9, 43.3) (Table 2).

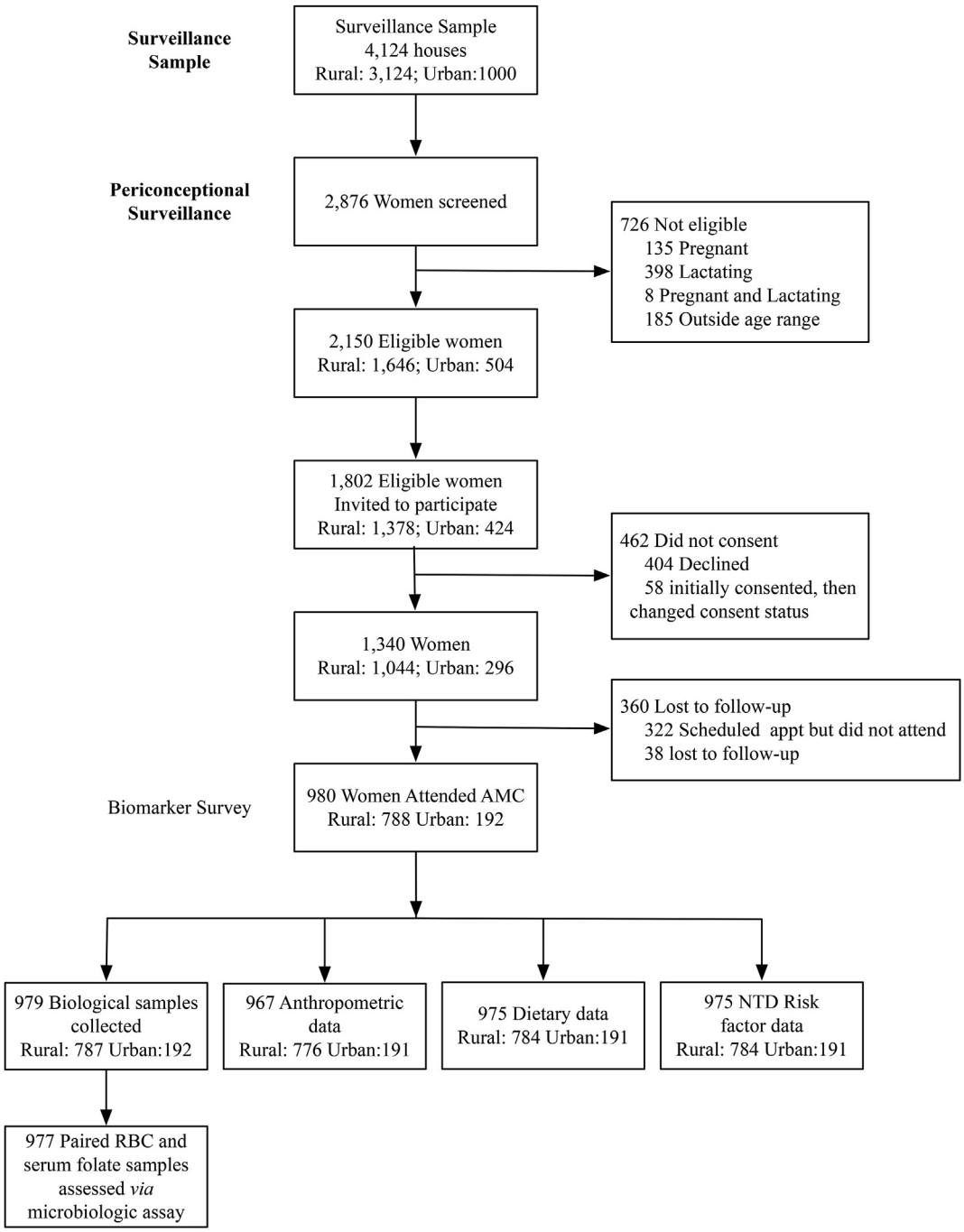

FIGURE 1. Participant flow chart.

Iron status.

The estimated serum folate insufficiency threshold did not differ substantially by anemia strata [Hb <12.0 g/dL vs. ≥12.0 g/dL; difference: -9.1 (-19.7, 0.4)]. However, the estimated threshold was lower in individuals with iron deficiency [BRINDA-adjusted serum ferritin <15.0 μg/L vs. ≥15.0 μg/L; difference: -10.6 (-28.1, -0.5) nmol/L] and in participants with low body iron index [body iron index <0.0 mg/kg vs. ≥0.0 mg/kg; difference: -12.3 (-27.5, -3.0) nmol/L] but did not substantially differ by iron insufficiency (i.e., serum ferritin <20.0 μg/L or <25.0 μg/L).

Vitamin B_{12} status. The estimated serum folate insufficiency threshold was 72.5 (52.1, 125.1) nmol/L in participants with vitamin B_{12} deficiency (<148 pmol/L), compared with 28.1 (25.7, 31.5) nmol/L in individuals who were not vitamin B_{12} deficient (≥148 pmol/L), with a 44.2 (23.6, 97.3) nmol/L difference between vitamin B_{12} deficiency strata (Table 2, Figure 3). The estimated threshold was 48.7 (41.0, 61.3) nmol/L in participants with vitamin B_{12} <221 pmol/L, compared with 24.3 (21.7, 27.8) nmol/L in individuals with vitamin B_{12} ≥221 pmol/L [vitamin B_{12} <221 pmol/L vs. vitamin B_{12} ≥221 pmol/L; difference: 24.3 (15.9, 37.5) nmol/L] (Table 2, Figure 3). The estimated threshold was also higher in participants with MMA >0.26 μmol/L [difference: 29.6 (18.3, 49.8) nmol/L], impaired vitamin B_{12} status [vitamin B_{12} <148 pmol/L and MMA >0.26 μmol/L vs. not; difference: 48.5 (24.1, 125.3) nmol/L], or cB_{12}

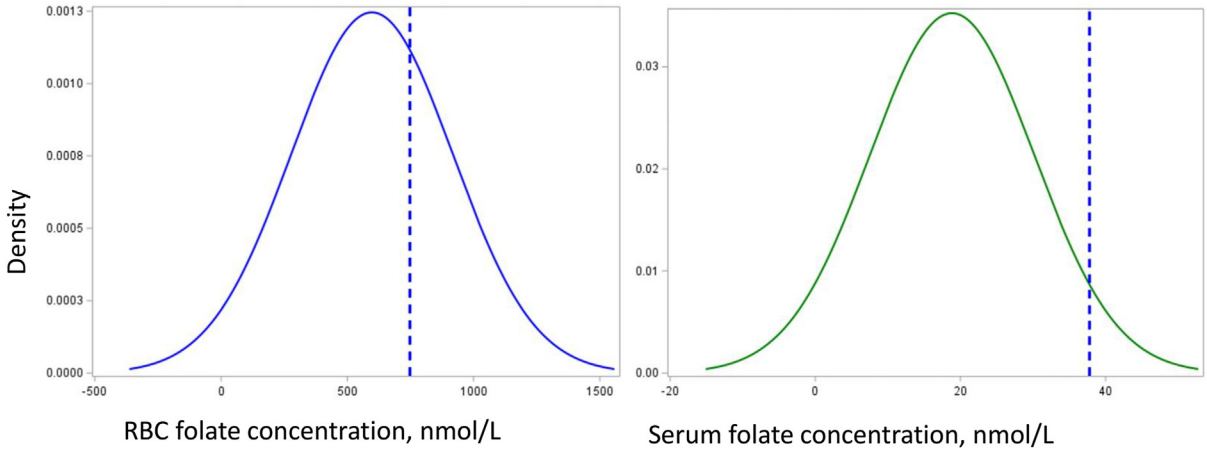

FIGURE 2. Serum folate and RBC folate distributions in the population, with corresponding insufficiency thresholds for optimal NTD prevention. RBC folate ($n = 977$, blue curve) and serum folate ($n = 977$, green curve) concentrations for this population. The blue dashed lines correspond to the RBC folate threshold for optimal NTD prevention and the corresponding estimated serum folate insufficiency threshold. Abbreviations: NTD, neural tube defect.

<-0.5 [cB_{12} <-0.5 vs. cB_{12} ≥-0.5; difference: 28.9 (21.3, 40.1) nmol/L].

Elevated homocysteine. The estimated threshold was higher in participants with Hcy >10.0 μmol/L [difference: 79.7 (25.3, 451.3) nmol/L] (Table 2).

Metabolic characteristics.

The estimated serum folate insufficiency thresholds by metabolic risk factors are presented in Table 2.

Body mass index. In analyses in adults (≥18 y), the estimated threshold did not substantially differ across BMI categories or by overweight strata (Table 2, Figure 4).

Central adiposity. In analyses in adults, the estimated serum folate insufficiency threshold was 10.5 nmol/L lower in participants with elevated WC (WC >88.9 cm vs. ≤88.9 cm; difference: -10.5 nmol/L (-17.6, -2.4) nmol/L] (Table 2, Figure 4). The serum folate insufficiency threshold did not substantially differ by elevated waist-hip ratio, elevated whole body fat mass, or elevated trunk fat mass (Table 2).

Elevated HbA1c. The estimated serum folate insufficiency threshold was lower in participants with elevated HbA1c, defined as HbA1c ≥6.5% [≥6.5% vs. <6.5%; difference: -19.5 (-27.2, -13.0) nmol/L] or HbA1c ≥5.7% [≥5.7% vs. <5.7%; difference: -15.5 (-25.9, -7.5) nmol/L] (Table 2, Figure 4).

Sensitivity analyses.

In analyses among participants with vitamin B_{12} ≥148 pmol/L, the estimated serum folate insufficiency threshold was 28.1 (25.6, 31.6) nmol/L. The estimated threshold was higher in participants with elevated MMA [MMA >0.26 μmol/L vs. ≤0.26 μmol/L; difference: 14.4 (6.8, 29.5) nmol/L; MMA >0.37 μmol/L vs. ≤0.37 μmol/L; difference: 17.7 (7.8, 41.8) nmol/L] or elevated Hcy [>10.0 μmol/L vs. ≤10.0 μmol/L; difference: 24.4 (2.5, 248.2) nmol/L], and lower in participants with elevated CRP [CRP >3.0 mg/L vs. ≤3.0 mg/L; difference: -7.8 (-13.3, -2.2) nmol/L] but did not vary substantially by age, anemia, iron deficiency, or other elevated inflammatory biomarkers (data not shown). In contrast, the serum folate insufficiency threshold was lower in participants with elevated WC [WC >88.9 vs. ≤88.9 cm; difference: -6.9 (-12.0, -1.6) nmol/L], elevated WHR [WHR ≥0.85 vs. <0.85; difference: -8.2 (-13.1, -3.1) nmol/L], elevated whole body fat [>35% vs. ≤35%; difference: -6.1 (-11.9, -0.8) nmol/L], and elevated trunk fat [>35% vs. ≤35%; difference: -6.5 (-12.1, -1.4) nmol/L]. The estimated threshold was also lower in participants with HbA1c ≥6.5% [≥6.5% vs. <6.5%; difference: -9.5 (-15.9, -2.4) nmol/L] (data not shown).

In analyses among participants with vitamin B_{12} ≥221 pmol/L, the estimated serum folate insufficiency threshold was 24.3 (21.5, 28.2) nmol/L. This threshold was lower among participants with low body iron index [<0.0 mg/kg vs. ≥0.0 mg/kg; difference: -9.6 (-21.7, -2.9) nmol/L] and in participants with elevated WHR [WHR ≥0.85 vs. <0.85; difference: -7.7 (-14.1, -1.5) nmol/L], but did not vary by other sociodemographic, nutritional, or metabolic characteristics.

Multivariable analyses.

The estimated serum folate insufficiency thresholds from multivariable analyses are summarized in Supplementary Tables 1–3. The overall serum folate insufficiency threshold was 36.7 (19.5, 68.6) nmol/L, after adjusting for Hb, CRP, and age. In multivariable analyses stratified by vitamin B_{12} status, the estimated serum folate insufficiency threshold was higher in participants with vitamin B_{12} <148 pmol/L [63.0 (26.3, 164.3) vs. 28.0 (17.0, 46.2) nmol/L] or vitamin B_{12} <221 pmol/L [46.1 (23.3, 89.9) vs. 24.2 (13.7, 44.3) nmol/L]. Findings were similar when models were adjusted for clusters identified through unsupervised hierarchical clustering methods.

Discussion

Monitoring RBC folate concentrations in women of reproductive age at the population level for optimal NTD prevention is recommended by the WHO. However, most countries monitor serum/plasma folate concentrations [23]; the WHO recommended that the relationship between serum folate and RBC folate concentrations be established at the population level to inform a corresponding serum folate threshold for optimal NTD prevention [22]. In this Bayesian analysis of population-based biomarker data from Southern India, the serum folate threshold was similar to the previously published threshold (24.3 vs. 25.6 nmol/L) among participants with sufficient vitamin B_{12} status. However, this threshold was more than 2-fold higher in individuals with vitamin B_{12} deficiency and substantially higher across other indicators of insufficient vitamin B_{12} status (<221 pmol/L, elevated MMA, cB_{12}, impaired vitamin B_{12} status) and elevated Hcy but lower in participants with elevated HbA1c. These findings suggest that a serum folate threshold for NTD

TABLE 2
Estimated serum folate insufficiency threshold corresponding to the RBC folate threshold for optimal NTD prevention: Overall and by sociodemographic, biochemical, and metabolic characteristics[1].

		Serum folate, nmol/L	Difference
Variables	n[2]	Median (95% Credible Interval)	Median (95% Credible Interval)
Overall	977	37.8 (33.9, 43.3)	n/a
Sociodemographic characteristics			
Age, y			
<26	277	54.0 (41.8, 81.5)	21.9 (6.6, 49.8)
26–36	452	33.9 (29.7, 40.5)	2.0 (-7.9, 10.2)
≥36	248	31.9 (26.9, 41.1)	Reference
Iron status			
Hemoglobin[3], g/dL			
<12.0	405	32.8 (28.5, 40.2)	-9.1 (-19.7, 0.4)
≥12.0	572	42.0 (36.0, 51.4)	Reference
Serum ferritin[4], μg/L			
<15.0	599	34.4 (30.7, 39.8)	-10.6 (-28.1, -0.5)
≥15.0	376	45.2 (36.4, 62.1)	Reference
<20.0	709	36.4 (32.2, 42.6)	-4.4 (-19.1, 5.1)
≥20.0	266	40.8 (33.4, 55.8)	Reference
<25.0	790	37.3 (33.0, 43.2)	-1.8 (-18.9, 7.8)
≥25.0	185	39.1 (31.7, 55.9)	Reference
Body iron index[5], mg/kg			
<0.0	378	32.9 (29.2, 38.3)	-12.3 (-27.5, -3.0)
≥0.0	597	45.3 (37.4, 59.9)	Reference
Specifically, for serum ferritin, CRP, serum B12, MMA, cB12, and HbA1c			
Serum CRP [6], mg/L			
≤1.0	425	41.6 (35.6, 51.4)	Reference
>1.0 to ≤3.0	261	36.6 (30.7, 47.4)	-4.9 (-16.5, 7.3)
>3.0 to ≤5.0	123	28.8 (23.9, 38.4)	-12.7 (-23.7, -1.7)
>5.0	168	29.3 (23.4, 42.8)	-12.2 (-23.8, 1.8)
>1.0	552	33.4 (29.5, 39.4)	-8.1 (-19.2, 0.7)
≤1.0	425	41.5 (35.4, 52.0)	Reference
>3.0	291	29.5 (25.2, 36.9)	-10.3 (-18.3, -1.6)
≤3.0	686	39.8 (35.2, 46.8)	Reference
>5.0	168	29.3 (23.3, 43.7)	-8.8 (-17.0, 5.1)
≤5.0	809	38.1 (34.2, 43.7)	Reference
Serum α-1 acid glycoprotein[7], g/L			
>1.0	216	33.5 (26.9, 48.1)	-4.7 (-13.6, 10.6)
≤1.0	759	38.2 (34.1, 44.2)	Reference
Vitamin B_{12} status			
Serum vitamin B_{12}[8], pmol/L			
<148	472	72.5 (52.1, 125.1)	44.2 (23.6, 97.3)
≥148	505	28.1 (25.7, 31.5)	Reference
<221	727	48.7 (41.0, 61.3)	24.3 (15.9, 37.5)
≥221	250	24.3 (21.7, 27.8)	Reference
Plasma MMA[8], μmol/L			
>0.26	592	55.6 (44.8, 75.7)	29.6 (18.3, 49.8)
≤0.26	545	25.9 (23.5, 29.3)	Reference
>0.37	432	68.3 (50.1, 115.7)	39.9 (21.5, 87.6)
≤0.37	545	28.2 (25.7, 31.6)	Reference
Impaired vitamin B_{12} status[8]			
Vitamin B_{12} <148 pmol/L and MMA >0.26 μmol/L	370	78.0 (53.7, 154.9)	48.5 (24.1, 125.3)
Not impaired vitamin B_{12} status	607	29.3 (26.7, 32.8)	Reference
Plasma homocysteine[9], μmol/L			
>10.0	157	114.9 (60.3, 485.3)	79.7 (25.3, 451.3)
≤10.0	820	34.7 (31.2, 39.6)	Reference
cB_{12}[8]			
≥1.5	20	14.6 (9.3, 22.4)	-14.7 (-21.5, -6.5)
-0.5 to <1.5	517	29.4 (26.6, 33.0)	Reference
-1.5 to <-0.5	333	54.2 (46.0, 66.5)	24.9 (17.6, 35.5)
-2.5 to <-1.5	100	70.7 (55.3, 94.6)	41.3 (26.9, 63.5)
<-2.5	7	123.2 (57.4, 280.1)	93.8 (28.8, 250.0)
≥-0.5	537	28.4 (25.8, 31.8)	Reference
<-0.5	440	57.4 (48.9, 70.0)	28.9 (21.3, 40.1)
BMI[10,11], kg/m^2			
<18.5	176	43.3 (34.4, 63.3)	9.5 (-1.5, 30.0)
18.5 to <25.0	440	33.7 (29.6, 39.9)	Reference
25.0 to <30.0	213	37.4 (29.4, 55.5)	3.7 (-6.4, 22.3)

(*continued on next page*)

TABLE 2 (*continued*)

		Serum folate, nmol/L	Difference
≥30.0	88	25.3 (21.1, 32.9)	-8.4 (-16.0, 0.0)
Waist circumference[10,12], cm			
>88.9	123	26.6 (22.6, 33.7)	-10.5 (-17.6, -2.4)
≤88.9	794	37.1 (33.1, 43.0)	Reference
Waist-hip ratio[10,12]			
≥0.85	191	30.5 (25.2, 41.0)	-5.6 (-13.2, 5.4)
<0.85	726	36.0 (32.3, 41.4)	Reference
Whole body fat[10,12], %			
>35	360	31.0 (27.1, 37.4)	-7.1 (-15.4, 0.8)
≤35	559	38.1 (33.3, 45.6)	Reference
Trunk fat[10,12], %			
>35	332	30.7 (26.7, 37.1)	-7.3 (-15.6, 0.7)
≤35	587	38.1 (33.4, 45.4)	Reference
HbA1c[13], %			
<5.7	732	43.8 (37.5, 53.7)	Reference
≥5.7	245	28.2 (24.9, 33.3)	-15.5 (-25.9, -7.5)
<6.5	928	40.5 (35.8, 47.3)	Reference
≥6.5	49	21.0 (17.8, 25.6)	-19.5 (-27.2, -13.0)

[1] Results outside assay LOD were set to ½ LOD (if below LOD) or 2 ×LOD (if above LOD); Results outside assay LODs: serum folate ($n = 1$ below LOD), vitamin B_{12} ($n = 4$ below LOD, $n = 6$ above LOD). BRINDA, Biomarkers Reflecting Inflammation and Nutritional Determinants of Anemia; cB_{12}, combined B_{12}; LOD, limit of detection; sTfR, soluble transferrin receptor; WHO, World Health Organization; 95% CI, 95% credible interval.

[2] Individuals contributing data to the Bayesian simulations by strata; reported results are summarizing the posterior distributions of 10,000 iterations

[3] Hemoglobin values from complete blood count; adjusted for smoking status [29]

[4] Serum ferritin concentrations were adjusted for inflammation via BRINDA methods [41]; cutoffs for iron deficiency (serum ferritin <15.0 μg/L) and iron insufficiency (serum ferritin <20.0 and <25.0 μg/L) from [33,34]

[5] Body iron index (mg/kg) was estimated using Cook's equation after adjusting serum ferritin for inflammation via BRINDA methods [41] and converting sTfR to Flowers Ramco equivalents (Ramco = 1.5 × Roche + 0.35) [35].

[6] CRP cutoffs for inflammation (>5.0 mg/L) and metabolic risk (>3.0 mg/L and >1.0 mg/L) from [33]

[7] α-1 acid glycoprotein cutoff (>1.0 g/L) based on WHO criteria [33]

[8] Cutoffs for vitamin B_{12} deficiency (<148 pmol/L) and vitamin B_{12} insufficiency (<221 pmol/L) from (30); cB_{12} was calculated using vitamin B_{12}, MMA, and homocysteine and categorized (i.e., ≥1.5, -0.5 to <1.5, -1.5 to <-0.5, -2.5 to <-1.5, <-2.5) using methods developed by Fedosov et al. [32] Elevated vitamin B12 (≥1.5), Adequate vitamin B12 (-0.5 to <1.5), Decreased vitamin B12 (-1.5 to <-0.5) , Possibly deficient (-2.5 to <-1.5) , Probably deficient (<-2.5) Sufficient vitamin B12 (-0.5 to >1.5), Deficient vitamin B12 (<-0.5).

[9] Homocysteine cutoffs (>15.0 and >10.0 μmol/L) from [6]

[10] Among adult participants (≥18 y; $n = 928$)

[11] BMI categories (<18.5, 18.5 to <25.0, 25.0 to <30.0, and ≥30.0 kg/m^2) as defined by the WHO [37]

[12] Anthropometric cutoffs (waist circumference >88.9 cm and waist-hip ratio ≥ 0.85) were defined based on WHO criteria [38]; elevated whole body (>35%) and trunk fat (>35%) were defined based on criteria suggested by the American Association of Clinical Endocrinology [39]

[13] HbA1c cutoffs (<5.7%, ≥5.7% to <6.5%, and ≥6.5%) were defined using the American Diabetes Association criteria [40]

prevention may be possible in some settings; however, it may not be appropriate in populations with a high prevalence of vitamin B_{12} insufficiency.

In this study, the serum folate insufficiency threshold for optimal NTD prevention was 44.2 nmol/L higher in individuals with vitamin B_{12} deficiency (i.e., 72.5 vs. 28.1 nmol/L). This is consistent with previous Bayesian analyses in China [24], which identified a higher plasma folate insufficiency threshold in participants with vitamin B_{12} <148 pmol/L. The magnitude of the threshold observed here was higher (i.e., 72.5 vs. 34.6 nmol/L), likely driven by much lower vitamin B_{12} concentrations and the higher prevalence of vitamin B_{12} deficiency (48.3 vs. 17.9%) [24]. The findings here were also consistent across all indicators of inadequate vitamin B_{12} status and cutoffs evaluated (i.e., <221 pmol/L, elevated MMA, cB_{12}, impaired vitamin B_{12} status). Vitamin B_{12} is required for the conversion of 5-MTHF to THF [6], maintenance of folates in RBCs, and retention of folates in proliferating cells such as developing RBCs during erythropoiesis [6]. Vitamin B_{12} deficiency can result in 5-MTHF accumulation, reduced THF availability, and impaired folate retention in RBCs [6]. Together, in the context of vitamin B_{12} deficiency, the estimated serum folate threshold for NTD prevention is above the physiological range of serum folate concentrations obtainable in the absence of high folic acid intake. Previous studies have found that folic acid intake does not impact the relationship between RBC folate and plasma folate thresholds [24], suggesting that this phenomenon (much higher serum threshold in vitamin B_{12} insufficiency) reflects differences in metabolism between folic acid and natural food folate and the importance of vitamin B_{12} status in folic acid naïve populations.

The estimated serum folate threshold did not vary substantially by overweight status or BMI strata. These findings contrast with previous analyses where the plasma folate threshold was lower in higher BMI strata [24]. Previous analyses were conducted in a population with lower prevalence of underweight and excluded women with vitamin B_{12} deficiency, constraining direct comparability of findings. Observational studies in nonpregnant women reported higher BMI was associated with lower serum folate but higher RBC folate concentrations [42]. The associations between higher BMI and lower serum folate concentrations may be due to increased blood volume or cellular folate uptake with increasing body size [43,44].

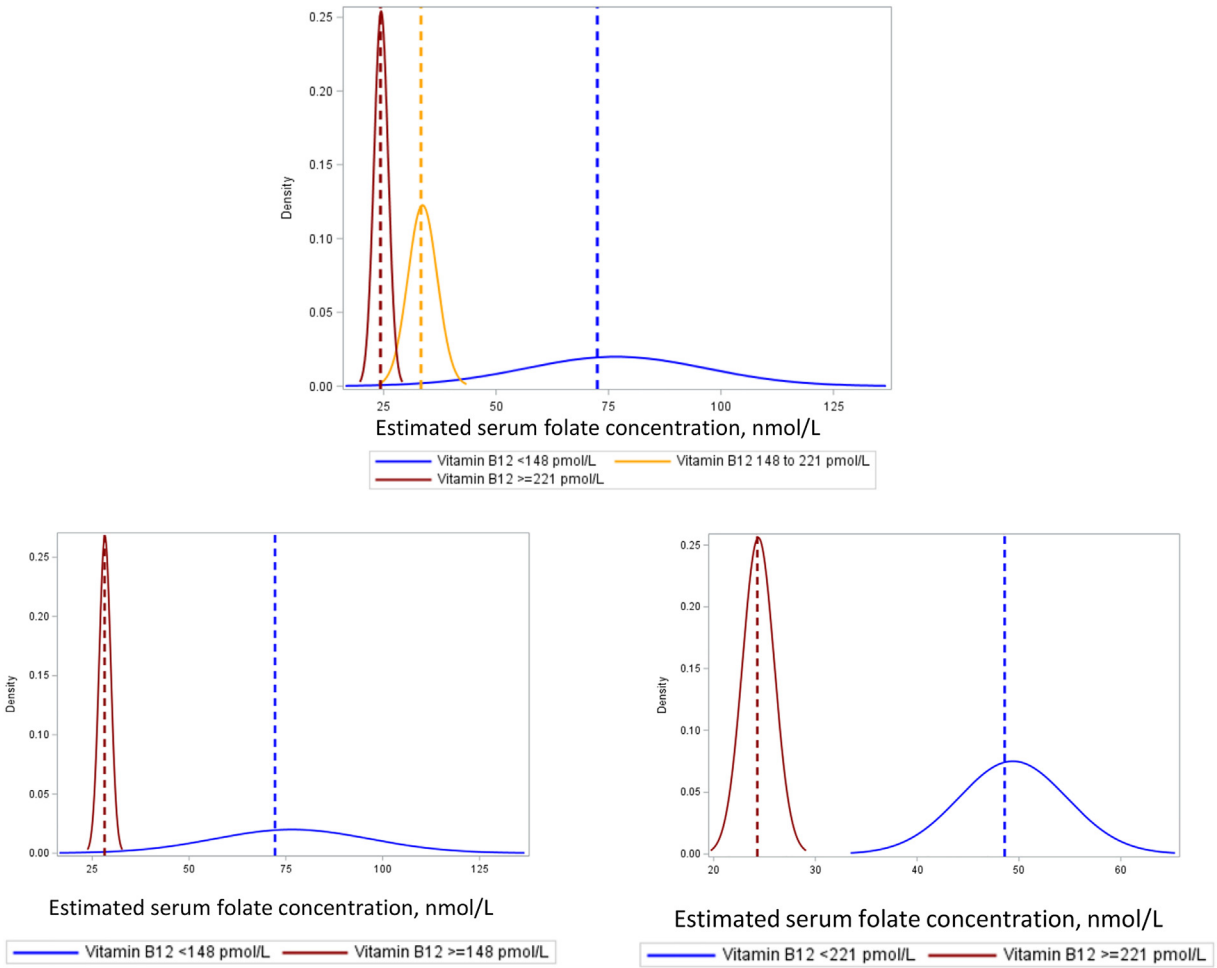

FIGURE 3. Estimated serum folate insufficiency threshold by vitamin B_{12} strata [n = 472 (<148 pmol/L), 255 (148–221 pmol/L), and 250 (≥221 pmol/L)]. Estimated serum folate concentrations corresponding to the RBC folate concentration of 748 nmol/L for optimal NTD prevention, by vitamin B_{12} status. The dashed lines correspond to the estimated median. Abbreviations: NTD, neural tube defect.

In the current study, the serum folate insufficiency threshold was lower in participants with elevated central adiposity or HbA1c (≥5.7% or ≥6.5%). HbA1c is an indicator used to screen for diabetes mellitus or to monitor glycemic control in individuals with diabetes. In randomized trials, folic acid supplementation was associated with lower insulin levels and insulin resistance [45,46], although it did not reduce HbA1c [45,46] or incidence of diabetes [47].

The serum folate threshold was also higher in individuals with insufficient (<221 pmol/L) or marginal (≥148 to <221 pmol/L) vitamin B_{12} status, suggesting that vitamin B_{12} may impact the serum folate threshold even in the absence of vitamin B_{12} deficiency. Evaluation of vitamin B_{12} status is important for estimation of the serum folate insufficiency threshold in populations with high prevalence of vitamin B_{12} deficiency and may be warranted in populations where a substantial proportion of the population has insufficient (<221 pmol/L) or marginal (≥148 to <221 pmol/L) vitamin B_{12} status. In the United States, although the prevalence of vitamin B_{12} deficiency in the overall population is 2.9% [48], it is estimated that 13.3% of nonpregnant women and 21.3% of pregnant women have vitamin B_{12} <221 pmol/L [49]. In other countries, nearly half of women have vitamin B_{12} <221 pmol/L, with ~30% ≥148 to <221 pmol/L [50–52].

In this analysis, metabolic risk factors (e.g., elevated HbA1c) impacted the serum folate threshold; however, the magnitude was approximately 8 to 10 nmol/L compared with 40.0 nmol/L observed by vitamin B_{12} deficiency strata. The impact of metabolic characteristics on this threshold is of interest etiologically, and additional studies exploring these associations would enhance the findings here. The impact of vitamin B_{12} status is immediately important when considering a serum folate threshold to monitor optimal NTD prevention at the population level.

Folic acid fortification policy is based on evidence from randomized supplementation trials [9–11] and population-based fortification programs [16–21] and is among the most efficacious and cost-effective public health interventions [53,54]. Folic acid fortification has been scaled up in over 70 countries and has been consistently associated with reduced NTDs, including populations with high prevalence of vitamin B_{12} deficiency [13]. The WHO recommends monitoring RBC folate at the population level; higher RBC folate concentrations have been associated with lower NTD risk in observational studies [8,14] and Bayesian analyses [15], and established an RBC threshold of ≥906 nmol/L for optimal NTD prevention [22]. The association between decreasing RBC folate concentrations and increasing NTD risk is independent of vitamin B_{12} [15].

The findings from the current study and Bayesian analyses of population data in China [24] suggest it may be possible to establish a serum/plasma folate threshold corresponding to the RBC threshold for

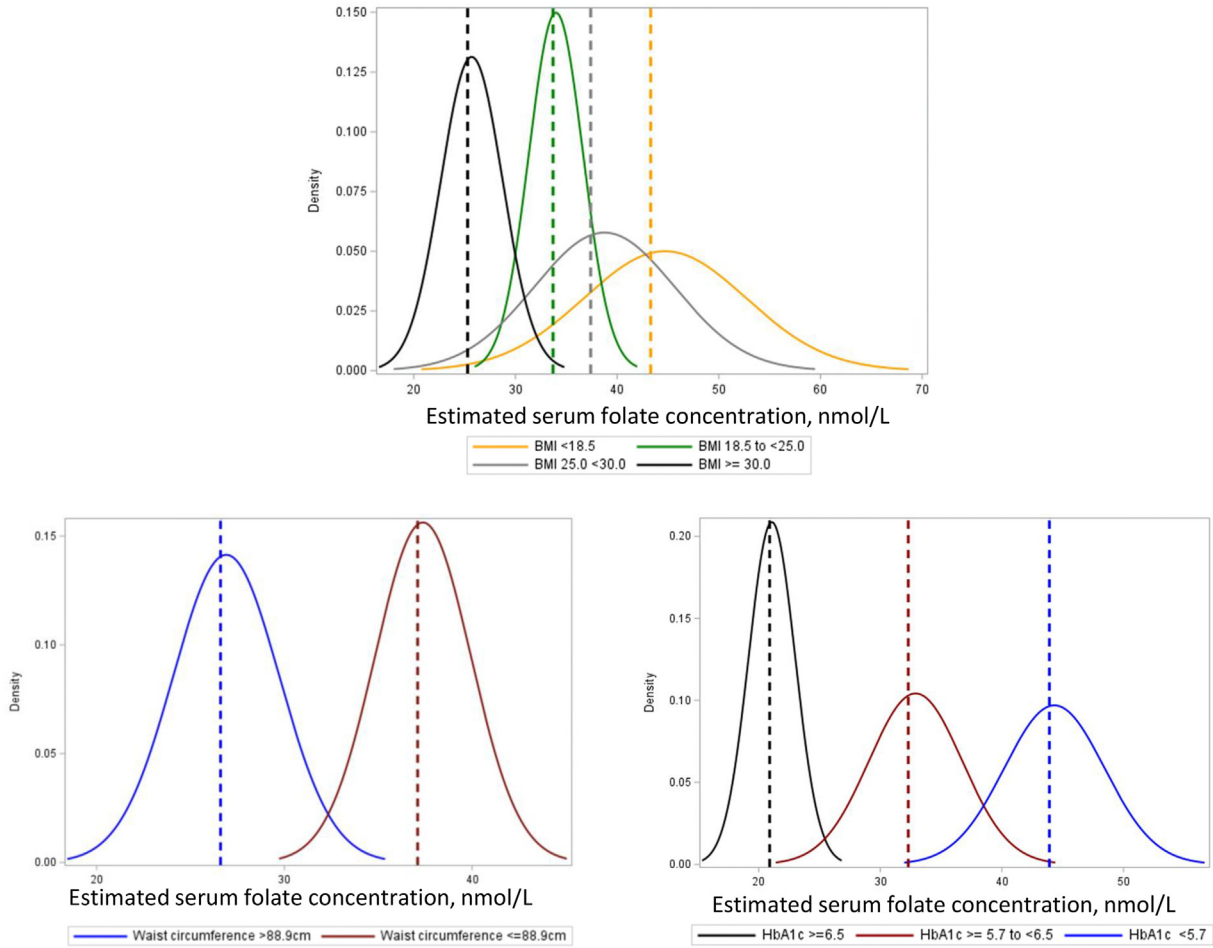

FIGURE 4. Estimated serum folate insufficiency threshold by BMI, waist circumference, and elevated HbA1c strata Estimated serum folate concentrations corresponding to the RBC folate concentration of 748 nmol/L, by BMI [n =176 (<18.5 kg/m^2), 440 (18.5 to <25.0 kg/m^2), 213 (25.0 to <30.0 kg/m^2), and 88 (≥30.0 kg/m^2)], waist circumference [n = 123 (>88.9 cm) and 794 (≤88.9 cm)], and elevated HbA1c strata [(n = 732 (<5.7%), 196 (≥5.7 to <6.5%), and 49 (≥6.5%)]. The dashed lines correspond to the estimated median. Abbreviations: HbA1c, glycated hemoglobin; NTD: neural tube defect.

NTD prevention. However, in both studies, the relationship between serum/plasma folate and RBC folate—and the corresponding thresholds for NTD prevention—were substantially modified by **vitamin B_{12} deficiency**. This impact of vitamin B_{12} deficiency on these folate biomarkers is consistent with the role of vitamin B_{12} in folate-mediated 1-carbon metabolism: vitamin B_{12} deficiency impairs conversion of folates to THF [6], retention of folates in RBCs during erythropoiesis, and maintenance of folates in RBCs [6]. Together, these findings suggest that although a serum folate threshold for NTD prevention may be estimated in the context of vitamin B_{12} sufficiency (>221 pmol/L), it may not be appropriate in populations with a high prevalence of vitamin B_{12} insuffieiency or deficiency, and serum/plasma folate concentrations should not be used to inform the design or monitoring of folic acid fortification programs. There is also emerging interest in vitamin B_{12} fortification, as vitamin B_{12} deficiency has been identified as an independent risk factor for NTDs [55] and can result in megaloblastic anemia [31] and pregnancy complications [56]. The efficacy of vitamin B_{12} fortification awaits appropriate randomized trials.

This study leverages a well-characterized population at high-risk for folate insufficiency and NTDs. This study applied previously developed Bayesian models [24] to estimate a serum folate threshold in a population with a high prevalence of vitamin B_{12} deficiency and included additional biomarkers of vitamin B_{12} status and metabolic outcomes.

This study has several important limitations. Its cross-sectional study design does not enable evaluation of the effects of risk factors on the estimated serum folate threshold. Although this study included participants who were not currently pregnant or lactating, the findings may not reflect the periconceptional period. The current study evaluated serum folate, whereas the previous Bayesian analysis evaluated plasma folate. Although these biomarkers are generally considered interchangeable, there may be unknown important differences. Assessment of the *MTHFR* genotype would improve comparability to the previous analysis, which identified a higher threshold with the *MTHFR TT* genotype [24]. However, previous studies have reported a low prevalence of *MTHFR* polymorphisms in South Indian populations [57], and modeling studies demonstrated that the RBC folate threshold is not influenced by the *MTHFR* genotype [15]. In this study, vitamin B_{12} deficiency and insufficiency were evaluated via total vitamin B_{12}, and additional functional vitamin B_{12} biomarkers, elevated MMA acid and Hcy, were examined. Although total vitamin B_{12} and MMA are specific indicators of vitamin B_{12} status, Hcy is influenced by a number of factors within (e.g., folate) [56] and outside (e.g., renal disease) [58] 1-carbon metabolism. The different circulating (i.e., total vitamin B_{12},

holo-transcobalamin) and functional (i.e., MMA, Hcy) biomarkers and cutoffs used pose challenges for evaluating vitamin B_{12} status and interpreting findings across different studies [30,31]. However, the findings here are consistent with the previous population-based study in China and were consistent across all vitamin B_{12} biomarkers and cutoffs evaluated. Inclusion of holo-transcobalamin in this study would improve assessment of vitamin B_{12} status and provide additional insights on the impact of vitamin B_{12} status on the estimated serum folate threshold. Inclusion of other metabolic risk factors (e.g., impaired kidney function [58]) also warrants consideration for inclusion in future analyses. The reported associations between serum folate and RBC folate are limited to the microbiologic assay and are not generalizable to other assays for folate assessment.

In our analysis of population-based biomarker data, the estimated serum folate threshold for optimal NTD prevention was consistent with the previous reports among participants with vitamin B_{12} >221 pmol/L. However, the serum folate threshold was substantially higher in individuals with vitamin B_{12} deficiency across all other indicators of insufficient vitamin B_{12} status and the cutoffs evaluated, suggesting a complex interaction between these two different measures of folate status in a folic acid naïve population. These findings suggest that although it may be possible to estimate a serum folate threshold for NTD prevention in some settings, it may not be appropriate in populations with high prevalence of vitamin B_{12} insufficiency.

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Author disclosures

The authors have no conflicts of interest to disclose related to this manuscript. Findings and conclusions in this report are those of the authors and do not necessarily represent the official positions of the Centers for Disease Control and Prevention, the DRDC, Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD), National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), or the National Institutes of Health.

Author Contributions

Analyses were replicated by Results Reproduction (R-squared), an independent statistical group at Cornell University. The authors' responsibilities were as follows—JLF, KSC: designed the research; AF: wrote the initial draft manuscript; KSC, CER, JLF: provided expert guidance on the statistical analyses and revised the manuscript; WB, CBJ, AF: supervised data collection and field activities; BB: conducted the laboratory analyses with expert guidance from CMP, MZ, SJ; AF: conducted the statistical analyses; JLF, KSC: had primary responsibility for the final content; and all authors: contributed to the development of the manuscript, provided feedback, and read and approved the final manuscript.

Data Availability

Data described in this manuscript, code book, and analytic code will be made available upon request at: https://archive.ciser.cornell.edu.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ajcnut.2023.01.016.

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